Order-Continuous Lattice Case of Cwikel's Plus-Minus Isometry Conjecture

Run `fa_banach_001`, agent `agent_lane_04`

2026-06-23

Status

Candidate partial result, likely valid. This packet proves the isometry conjectured in Remark 4.1 of Cwikel's paper [2] for couples of complex Banach lattices of measurable functions whose endpoint lattice norms are order continuous. The unrestricted case of arbitrary complexified Banach lattices is not claimed here.

Source Problem

The source paper is Michael Cwikel, "Some alternative definitions for the 'plus-minus' interpolation spaces $\langle A_0, A_1\rangle_\theta$ of Jaak Peetre," arXiv:1502.00986v2 [2]. The relevant statement is Remark 4.1 on page 13.

"periodic" norms on $[A_0, A_1]_\theta$.

Remark 4.1. The inclusion $\langle A_0, A_1\rangle_\theta \subset [A_0, A_1]_\theta$ is the easy part of the proof of another interesting property of the space $\langle A_0, A_1\rangle_\theta$, namely that it *coincides* with $[A_0, A_1]_\theta$, to within equivalence of norms, whenever (A_0, A_1) is a couple of complexified Banach lattices of measurable functions on the same underlying measure space. This property has sometimes proved useful, for example in [6], and I hope to use it further in the very near future, in a forthcoming paper (which in fact has been the main motivation for me to write this note). I also conjecture that this coincidence to within equivalence of norms of $\langle A_0, A_1\rangle_\theta$ and $[A_0, A_1]_\theta$ for couples of lattices is even an *isometry* when $\langle A_0, A_1\rangle_\theta$ is equipped with the norm of $\langle A_0, A_1\rangle_{\theta,(0)}$, i.e., the norm associated with the apparently new "continuous" method for its construction presented in Section 3. Hence my interest in explicitly noting that the embedding $\langle A_0, A_1\rangle_{\theta,(r)} \subset [A_0, A_1]_\theta$ has norm one.

The remark recalls that, for couples of complexified Banach lattices of measurable functions on the same measure space, the plus-minus space $\langle A_0, A_1\rangle_\theta$ is known to coincide with Calderon's complex interpolation space $[A_0, A_1]_\theta$ up to equivalent norms. It then conjectures that the coincidence is isometric when the plus-minus side is equipped with the continuous norm $\langle A_0, A_1\rangle_{\theta,(0)}$.

Idea of the Proof

Cwikel already proves the norm-one inclusion $\langle X_0, X_1\rangle_{\theta,(0)} \subset [X_0, X_1]_\theta$, so only the reverse norm inequality is needed. For Banach lattices, the complex interpolation norm is given by the Calderon product norm. A Calderon factorization can be written in the pointwise form

$$|f| \le b, \qquad be^{-\theta s} \in X_0, \qquad be^{(1-\theta)s} \in X_1.$$

The scalar parameter s says where the mass of f should be placed on the interpolation line. Put a narrow kernel in the t-variable centered at $s(\omega)$:

$$u(t,\omega) = \alpha(\omega)b(\omega)k(t-s(\omega)), \qquad |\alpha| \le 1, \quad \alpha b = f.$$

Then $\int u = f$, while testing against every $|\phi| \le 1$ gives the pointwise endpoint estimates

$$\left|\int e^{-\theta t}\phi(t)u(t)\,dt\right| \lesssim be^{-\theta s}, \qquad \left|\int e^{(1-\theta)t}\phi(t)u(t)\,dt\right| \lesssim be^{(1-\theta)s}.$$

For finite-valued s this is directly an admissible continuous plus-minus representation. For general s, truncate to bounded s-bands and quantize. Order continuity of the two endpoint lattice norms makes the discarded tails small in both endpoint estimates, which lets the finite-band argument pass to the limit.

Definitions

Let X_0 and X_1 be complex Banach lattices of measurable functions on a common measure space, and let $0 < \theta < 1$. We write $\langle X_0, X_1\rangle_{\theta,(0)}$ for Cwikel's continuous plus-minus space and norm from [2, Section 3]. Thus an element f is represented as $f = \int_{-\infty}^{\infty} u(t)\,dt$, where u is a locally piecewise continuous $X_0 \cap X_1$-valued function whose weighted improper integrals are unconditionally convergent after multiplication by all locally piecewise continuous scalar functions ϕ with $\|\phi\|_\infty \le 1$. Its J-norm is

$$\|u\|_J = \max_{j=0,1} \sup_{\phi} \left\| \int_{-\infty}^{\infty} e^{(j-\theta)t}\phi(t)u(t)\,dt \right\|_{X_j},$$

and $\|f\|_{\langle X_0,X_1\rangle_{\theta,(0)}}$ is the infimum of $\|u\|_J$ over such representations.

We use the Calderon product gauge in the equivalent form

$$\|f\|_{X_0^{1-\theta}X_1^{\theta}} = \inf M,$$

where the infimum is over all measurable $b \ge |f|$ and real measurable s such that

$$be^{-\theta s} \in X_0, \quad be^{(1-\theta)s} \in X_1, \quad \left\|be^{-\theta s}\right\|_{X_0} \le M, \quad \left\|be^{(1-\theta)s}\right\|_{X_1} \le M.$$

This is the usual Calderon product norm, after the standard balancing of the two endpoint factors.

A Kernel Lemma

Lemma 1 (Finite logarithmic-ratio bands)**.** *Let g be a measurable function. Suppose that for some measurable $b \ge |g|$ and some real measurable s taking values in a bounded interval,*

$$y_0 := be^{-\theta s} \in X_0, \qquad y_1 := be^{(1-\theta)s} \in X_1.$$

Then

$$\|g\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{\|y_0\|_{X_0}, \|y_1\|_{X_1}\}.$$

Proof. First assume that s takes only finitely many values. Put $\alpha = g/b$ on $\{b > 0\}$ and $\alpha = 0$ on $\{b = 0\}$, so that $|\alpha| \le 1$ and $\alpha b = g$. Let k_δ be the normalized indicator of $[-\delta/2, \delta/2]$. Define

$$u_\delta(t,\omega) = \alpha(\omega)b(\omega)k_\delta(t - s(\omega)).$$

Because s has finite range, u_δ is locally piecewise constant as an $X_0 \cap X_1$-valued function, has compact t-support, and is therefore admissible in Cwikel's continuous J-space. Also

$$\int_{-\infty}^{\infty} u_\delta(t,\omega)\,dt = g(\omega).$$

For every admissible scalar test function ϕ with $|\phi| \le 1$, pointwise almost everywhere,

$$\left| \int e^{-\theta t}\phi(t)u_\delta(t)\,dt \right| \le be^{-\theta s} \int e^{-\theta r}k_\delta(r)\,dr = C_{0,\delta}y_0,$$

and similarly

$$\left| \int e^{(1-\theta)t}\phi(t)u_\delta(t)\,dt \right| \le C_{1,\delta}y_1,$$

where

$$C_{0,\delta} = \int e^{-\theta r}k_\delta(r)\,dr, \qquad C_{1,\delta} = \int e^{(1-\theta)r}k_\delta(r)\,dr.$$

Both constants tend to 1 as $\delta \downarrow 0$. The lattice property gives

$$\|u_\delta\|_J \le \max\{C_{0,\delta}\|y_0\|_{X_0}, C_{1,\delta}\|y_1\|_{X_1}\}.$$

Letting $\delta \downarrow 0$ proves the finite-valued case.

Now assume only that s is bounded. For every $\eta > 0$, choose a finite-valued measurable q with $|q - s| \le \eta$. Applying the finite-valued case with q in place of s gives

$$\|g\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{\left\|be^{-\theta q}\right\|_{X_0}, \left\|be^{(1-\theta)q}\right\|_{X_1}\} \le e^\eta \max\{\|y_0\|_{X_0}, \|y_1\|_{X_1}\}.$$

Letting $\eta \downarrow 0$ proves the lemma. □

Partial Isometry Theorem

Theorem 1. *Let X_0 and X_1 be complex Banach lattices of measurable functions on the same measure space. Assume that both endpoint lattice norms are order continuous. Then, for every $0 < \theta < 1$,*

$$\langle X_0, X_1\rangle_{\theta,(0)} = [X_0, X_1]_\theta$$

isometrically.

Proof. Cwikel proves the contractive embedding

$$\|f\|_{[X_0,X_1]_\theta} \le \|f\|_{\langle X_0,X_1\rangle_{\theta,(0)}}$$

for every Banach couple [2, Theorem 4.3]. It remains to prove the reverse inequality in the present lattice setting.

Under the order-continuity hypothesis, the Calderon product $X_0^{1-\theta}X_1^\theta$ gives the complex interpolation space isometrically; this is the standard Calderon-Lozanovskii representation theorem for Banach lattices, in the order-continuous case [1, 3]. It is therefore enough to prove

$$\|f\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \|f\|_{X_0^{1-\theta}X_1^\theta}.$$

Fix $M > \|f\|_{X_0^{1-\theta}X_1^\theta}$. Choose $b \ge |f|$ and real measurable s such that

$$y_0 = be^{-\theta s} \in X_0, \qquad y_1 = be^{(1-\theta)s} \in X_1,$$

and

$$\|y_0\|_{X_0} \le M, \qquad \|y_1\|_{X_1} \le M.$$

For $N < K$, set

$$f_{N,K} = f\,\mathbf{1}_{\{N<|s|\le K\}}.$$

Lemma 1, applied to the bounded s-band $\{N < |s| \le K\}$, gives

$$\|f_{N,K}\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{\left\|y_0\mathbf{1}_{\{N<|s|\le K\}}\right\|_{X_0}, \left\|y_1\mathbf{1}_{\{N<|s|\le K\}}\right\|_{X_1}\}.$$

Since the endpoint norms are order continuous and the sets $\{|s| > N\}$ decrease to the empty set, the right hand side tends to 0 as $N \to \infty$, uniformly in $K > N$.

Consequently the net $f\,\mathbf{1}_{\{|s|\le K\}}$ is Cauchy in the continuous plus-minus norm: if $K_2 > K_1$, their difference is supported on $K_1 < |s| \le K_2$ and the preceding estimate applies. Cwikel proves that the continuous plus-minus space is complete [2, Section 5]; moreover it is continuously embedded in $X_0 + X_1$. The same truncations converge to f in $X_0 + X_1$: on $\{s > K\}$ we have $|f| \le b = e^{-(1-\theta)s}y_1 \le e^{-(1-\theta)K}y_1$, while on $\{s < -K\}$ we have $|f| \le b = e^{\theta s}y_0 \le e^{-\theta K}y_0$. Hence the $X_0 + X_1$-norm of the discarded part is at most $e^{-\theta K}\|y_0\|_{X_0} + e^{-(1-\theta)K}\|y_1\|_{X_1}$. Therefore

$$f = \lim_{K\to\infty} f\,\mathbf{1}_{\{|s|\le K\}}$$

belongs to $\langle X_0, X_1\rangle_{\theta,(0)}$.

For each fixed K, Lemma 1 gives

$$\left\|f\,\mathbf{1}_{\{|s|\le K\}}\right\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \max\{\left\|y_0\mathbf{1}_{\{|s|\le K\}}\right\|_{X_0}, \left\|y_1\mathbf{1}_{\{|s|\le K\}}\right\|_{X_1}\} \le M.$$

Passing to the limit in the Banach space $\langle X_0, X_1\rangle_{\theta,(0)}$ yields $\|f\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le M$. Since $M > \|f\|_{X_0^{1-\theta}X_1^\theta}$ was arbitrary, $\|f\|_{\langle X_0,X_1\rangle_{\theta,(0)}} \le \|f\|_{X_0^{1-\theta}X_1^\theta} = \|f\|_{[X_0,X_1]_\theta}$. Together with Cwikel's contractive embedding, this proves the asserted isometry. □

Limitations

The proof uses order continuity exactly when it removes the tails $\{|s| > N\}$ in both endpoint lattice norms. It does not settle Cwikel's full conjecture for arbitrary complexified Banach lattices; in particular, endpoint spaces with non-order-continuous norms such as L^∞-type lattices require a different argument or an obstruction.

Verification Notes

The proof was checked against the definitions and results in [2]. The key finite-band construction is pointwise and uses only the lattice domination property. The only external theorem used is the standard Calderon-Lozanovskii identification of complex interpolation with the Calderon product under the stated order continuity hypothesis.

Bounded novelty check performed on 2026-06-23: the run indexes were searched for `1502.00986`, `plus-minus`, `Peetre`, `Calderon`, and lattice-isometry keywords; no existing packet for this arXiv id or this order-continuous subcase was found. Web searches for exact phrases from Remark 4.1 and for "plus-minus interpolation Banach lattices complex interpolation isometry" did not reveal a later exact resolution of the conjecture.

Recommended verifier focus: check the passage from bounded s-bands to the unbounded measurable s using completeness of Cwikel's continuous plus-minus space, and confirm that the cited Calderon-Lozanovskii isometric representation is available under the endpoint order continuity hypotheses as stated.

References

[1] A. P. Calderon, Intermediate spaces and interpolation, the complex method, *Studia Math.* 24 (1964), 113–190.

[2] M. Cwikel, Some alternative definitions for the "plus-minus" interpolation spaces $\langle A_0, A_1\rangle_\theta$ of Jaak Peetre, arXiv:1502.00986v2, 2015.

[3] G. Ya. Lozanovskii, On some Banach lattices, *Sibirsk. Mat. Zh.* 10 (1969), 584–599.